INCH-POUND

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COMMERCIAL ITEM DESCRIPTION

THREAD, LINEN

The General Services Administration has authorized the use of this commercial item description (CID) by all federal agencies.

1. SCOPE. This document covers linen thread for sewing by machine or hand.

2. CLASSIFICATION. The thread shall be in the following types.

2.1 Types.

Type I - Three cord (stitching) thread. Natural finish having no consistency or luster other than that inherent in the linen. Polished and lustrous, having a smooth dressed surface with no protruding fibers.

Type II - Lockstitch twist thread. Natural finish having no consistency or luster other than that inherent in the linen. Polished and lustrous, having a smooth dressed surface with no protruding fibers.

Type III - Hand Shoe Thread. Natural Finish, having no consistency or luster other than that inherent in the linen.

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FSC 8310

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3. SALIENT CHARACTERISTICS.

3.1 Description. The thread shall be made of "long-line" flax. Types I and II shall have "S" or "Z" final twist as specified in the applicable end item specification or in the contract or order. Type III shall have a "Z final twist.

3.2 Color. The thread shall be natural color (boiled or scoured), bleached white, or as specified in the contract or purchase order. The finished thread shall be a good (good is defined as a perceptible not appreciable difference in color) match to the standard sample. If no standard sample is available, the thread shall be a good match to the end item base cloth.

3.2.1 Visual shade matching. The color and appearance of the material shall match the standard sample when viewed using the AATCC EP9, Evaluation Procedure for Visual Assessment of Color Difference of Textiles, Option C (see 7.4), with sources simulating artificial daylight D_{65} or D_{75} illuminant with a color temperature of 6500 (± 200) K or 7500 (± 200) K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2856 (± 200) K.

3.2.2 Colorfastness. The colorfastness of dyed finished thread shall be as specified in the applicable end item specification or as set forth in the contract except that when requirements are not stated or referenced to a standard sample for colorfastness, the thread shall conform to the requirements listed in Table I.

Table I. Colorfastness requirements.

Characteristic	Requirement	Test method
Colorfastness to laundering, (color change only)(min.)	3-4	AATCC TM61, Test 4A, 1/
Colorfastness to light, (after 40 AFU or 170 kJ/(m2nm) @420 nm), (min.) 2/	3-4	AATCC TM16.3, 1/

1/ Rated using AATCC EP1, Evaluation Procedure for Gray Scale for Color Change.
2/ AFU- AATCC Fading Units.

3.3 Physical requirements of thread. The thread shall conform to the physical requirements specified in Table II.

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TABLE II Physical requirements.

	Tex Natural	Tex Polished	Breaking Strength (lbs.) (min.)	Test Method
Three Cord Stitching Thread	90	105	6.0	ASTM D204 1/, 2/
	120	135	7.5	ASTM D204 1/, 2/
	150	150	8.5	ASTM D204 1/, 2/
	180	180	10.0	ASTM D204 1/, 2/
	210	210	12.0	ASTM D204 1/, 2/
	240	270	18.0	ASTM D204 1/, 2/
	300	350	19.0	ASTM D204 1/, 2/
Lockstitch Twist Thread	240	270	18.0	ASTM D204 1/, 2/
	350	350	24.0	ASTM D204 1/, 2/
	400	450	30.0	ASTM D204 1/, 2/
	500	500	38.0	ASTM D204 1/, 2/
	600	600	44.0	ASTM D204 1/, 2/
	700	700	52.0	ASTM D204 1/, 2/
	800	800	59.0	ASTM D204 1/, 2/
	800	900	65.0	ASTM D204 1/, 2/
	900	900	72.0	ASTM D204 1/, 2/
	1000	1000	81.0	ASTM D204 1/, 2/
Hand ShoeThread	120	N/A	6.0	ASTM D204 1/, 2/
	150		10.0	ASTM D204 1/, 2/

1/ Testing speed shall be 12 (± 0.5) in./min., and a 10-inch gauge length shall be used.
2/ Five (5) determinations per sample unit.

3.4 Labile sulfur. The use of dyes and compounds containing sulfur capable of oxidation to sulfuric acid is prohibited. The thread shall be “Free” of labile sulfur when tested as specified in 3.4.1.

3.4.1 Presence of labile sulfur test. Two (2) 1.50 (± 0.01) gram samples from each sample unit shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. Prepare an acidic tin(II) chloride (stannous chloride) solution by dissolving 100 g SnCl2 (ACS grade) in 100 mL of 35% hydrochloric acid (ACS grade) and adding the mixture to 50 mL of distilled water. Prepare a lead acetate solution by dissolving 5.0 g of lead acetate (CP reagent grade) in a small amount of distilled water and diluting to a total of 100 mL. If the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear. The samples shall be submersed the acidic tin(II) chloride solution. A filter paper wet out with the lead acetate solution shall be placed over the top of the test tube. The test tube containing the textile sample, tin(II) chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

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Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

The rating shall be recorded. The results shall be recorded as "pass" or "fail.

3.5 Finishing materials. The thread shall have no chemical finishes or treatments other than those commonly used in commercial threads. No chemical finishes or treatments shall be applied for the purpose of increasing breaking strength.

3.6 Length and put-up. Unless otherwise specified, the thread shall be put-up on a nominal length per holder or commercial spools, cones, or tubes as specified in the contract. The thread shall be wound around the specified holder in one continuous piece, so that each turn and layer is free of entanglement. The outside ending of the thread shall be secured to prevent unwinding, loosening or slippage during handling, shipping or storage.

3.7 Labeling. Each holder shall have a label adhered securely to remain in place and be clearly legible until all thread has been removed. The label shall be printed and include information related to type, length in yards, direction of twist, color, tex, and name of thread manufacturer. In addition, it shall be labeled or ticked for fiber content in accordance with the Rules and Regulations under the Textile 16 CFR Fiber Products Identification Act (see 7.1.2).

3.8 Workmanship. The finished thread shall conform to the quality of the product established in this document. The thread shall be free from defects which would affect its serviceability, appearance, uniformity and quality assurance standards defined by the technical data in the bid package.

4. REGULATORY REQUIREMENTS. Unless otherwise specified, the offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403, 23.404(b) of the Federal Acquisition Regulation FAR, or the sustainable acquisition of EO 13693; as applicable. (See 7.2.4)

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The thread provided shall meet the salient characteristics of this commercial item description and conform to the producer's own specifications, standards and quality assurance practices. The Government reserves the right to require proof of such conformance.

5.2 Market acceptability. The thread offered or generic equivalent, must have been sold to the government or commercial market.

5.3 End item inspection procedures. Sampling for inspection shall be conducted in accordance with ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection of Attributes (see 7.2.3).

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5.4 Visual examination. The finished thread shall be examined for the defects listed in Table III as defined in ASTM D3990. The lot size shall be expressed in units of spools, cones, or tubes. The sample unit shall be one (1) spool, cone, or tube.

TABLE III. Visual examination defects.

Material defects: Any spot or stain on outside, discoloration or color not as specified, any outwardly visible part shaded, any serious material defects or objectionable odor.
Workmanship: Any component part omitted, twisted, not specified type and size. Failure to meet product conformance; serviceability, reliability and quality of materials as defined by this document.
Labels: Label(s) missing, incorrect, or illegible; bar code label causes damage to the item.
Packaging: Not packaged in accordance with the contract or purchase order.

5.5 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order (see 7.5).

6. PACKAGING. The preservation, packing and marking shall be as specified in the contract or order (see 7.5).

7. NOTES.

7.1 Source of Government documents:

7.1.1 Copies of Government documents are available online at https://quicksearch.dla.mil.

7.1.2 16 CFR Part 303 Rules and Regulations under the Textile Fiber Products Identification Act are available online at https://www.ftc.gov.

7.2 Sources of Non-government Documents.

7.2.1 AATCC Standards are available for purchase online at https://www.aatcc.org.

7.2.2 ASTM Standards are available for purchase online at https://www.astm.org.

7.2.3 ASQ/ANSI Z1.4 – Sampling Procedures are available online at https://www.asq.org.

7.2.4 EPA and EO 13693 information are available online at https://www.epa.gov.

7.2.5 Federal Acquisition Regulations (FAR) are available online at https://acquisition.gov .

7.3 Intended use. The thread is intended for sewing by machine or by hand.

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7.4 Visual shade matching. In 2017, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5º) angle relative to the horizontal. The observer will view the specimens at a 90º angle, relative to the plane of the specimens.".

7.5 Standard samples. For access to standard shade samples of thread, address the contracting activity issuing the invitation for bids or request for proposal.

7.6 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this commercial item description
b. Type, and size required (see 2.1), and (3.3)
c. Color required (see 3.2)
d. Product conformance provisions (see 5.1)
e. Acceptance criteria (see 5.5)
f. Packaging requirements (see 6.)

7.6 Key words.

Hand Shoe
Lockstitch twist
Three cord

MILITARY INTERESTS:

Custodians:
Army – GL
Navy - NU

CIVIL AGENCY COORDINATING ACTIVITY:
GSA — FAS

Preparing Activity
DLA – CT

(Project: 8310-2021-001)

Review Activities:

Army- MD, EA, CR4
Navy- MC
Other agency – DS

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